THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Life Processes in Plants** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** autotrophs	Auto means self.
2	**(c)** blue-black	Starch test.
3	**(c)** leguminous plants	They fix nitrogen.
4	**(c)** make food for all	Base of the food chain.
5	**(a)** photosynthesis	Photo means light.
6	**(a)** xylem	Xylem.
7	**(c)** fixing atmospheric nitrogen	Provide nitrates.
8	**(a)** plants	Plants make food.
9	**(b)** chlorophyll	It traps sunlight.
10	**(a)** phloem	Phloem.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Making of food by plants using sunlight (1); Carbon dioxide + water gives glucose + oxygen (1).	1+1
12	Xylem carries water and minerals (1); Phloem carries food (1).	1+1
13	Two organisms live together to benefit each other (1); Lichen or Rhizobium with legumes (1).	1+1
14	Carbon dioxide (1); Water and sunlight (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Carbon dioxide + water (1); In sunlight with chlorophyll (1); gives glucose + oxygen (1).	1+1+1
16	Leaves form a pitcher (1); Insects fall in (1); Enzymes digest them (1).	1+1+1
17	Fix nitrogen (1); Enrich soil (1); Reduce fertiliser need (1).	1+1+1
	OR: Cover part of a leaf (1); Test for starch after some hours (1); Only exposed part turns blue-black (1).	1+1+1
18	Alga makes food (1); Fungus gives shelter and water (1); Both benefit (1).	1+1+1
	OR: Photosynthesis makes food (1); Respiration releases energy (1); One in light, other always (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Carbon dioxide and water (1); Sunlight (1); Chlorophyll in chloroplasts (1); Glucose and oxygen (1); In leaves (1).	1+1+1+1+1
	OR: Take in oxygen (1); Give out carbon dioxide (1); Roots respire in soil air (1); Shed leaves (1); Store gums, resins (1).	1+1+1+1+1
20	Variegated leaf test (1); Covered leaf test (1); Leaf in alkali (CO_2 absorbed) (1); Iodine test (1); Only suitable leaf parts give starch (1).	1+1+1+1+1
	OR: Make food (1); Release oxygen (1); Absorb CO_2 (1); Give habitat (1); Prevent soil erosion (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that photosynthesis takes place only at night, or that respiration takes place only at night; photosynthesis needs **light**, respiration goes on **all the time**.
- Confusing xylem and phloem: **xylem** = water and minerals up, **phloem** = food from leaves.
- Writing that oxygen is a raw material of photosynthesis; the raw materials are carbon dioxide and water, and oxygen is the *by-product*.
- Mixing up transpiration (loss of water vapour through stomata) with absorption of water by the root hairs.
- Writing that Cuscuta makes its own food; it is a **parasite** without chlorophyll.
- Forgetting that iodine test shows **starch** (blue-black), not glucose, and that the leaf should first be boiled in alcohol to remove chlorophyll.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Life Processes in Plants** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** symbiosis	Both benefit.
2	**(a)** photosynthesis	Plants release.
3	**(a)** carbon dioxide and water	Plus sunlight.
4	**(d)** transpiration	Through stomata.
5	**(b)** stomata and roots	Gas exchange.
6	**(c)** guard cells	Control opening.
7	**(b)** glucose (starch)	Carbohydrate.
8	**(b)** cooling the plant and pulling water up	Suction force.
9	**(a)** air spaces in soil	For respiration.
10	**(d)** night	Less need for CO2.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Parasitic (1); Saprotrophic (1).	1+1
12	They make food (1); Others depend on them (1).	1+1
13	Traps sunlight (1); Needed for making food (1).	1+1
14	Tiny pores in leaves (1); Gas exchange and transpiration (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Root hairs absorb (1); Xylem carries upward (1); Transpiration pulls water (1).	1+1+1
16	Lives in root nodules (1); Fixes nitrogen (1); Makes nitrates (1).	1+1+1
17	Cools the plant (1); Pulls up water (1); Moves minerals (1).	1+1+1
	OR: Through stomata (1); Photosynthesis in the day (1); Gas exchange (1).	1+1+1
18	Cells around stomata (1); Open and close them (1); Control gas exchange (1).	1+1+1
	OR: No photosynthesis (1); No starch made (1); Iodine remains brown (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Autotrophic: green plants (1); Parasitic: Cuscuta (1); Saprotrophic: fungi (1); Insectivorous: pitcher plant (1); Symbiotic: lichen, Rhizobium (1).	1+1+1+1+1
	OR: Nitrogen is present in air (1); Rhizobium fixes it (1); Plants take nitrogen compounds from the soil (1); Animals eat plants (1); Decay returns nitrogen to the soil (1).	1+1+1+1+1
20	Roots absorb (1); Xylem carries upward (1); Phloem carries food (1); Transpiration pull (1); Stored in roots, fruits (1).	1+1+1+1+1
	OR: Glucose is the base (1); Nitrogen from nitrates (1); Fats made from glucose (1); Proteins from amino acids (1); Stored in seeds (1).	1+1+1+1+1

Mistakes to avoid:

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- Confusing xylem and phloem: **xylem** = water and minerals up, **phloem** = food from leaves.
- Writing that oxygen is a raw material of photosynthesis; the raw materials are carbon dioxide and water, and oxygen is the *by-product*.
- Mixing up transpiration (loss of water vapour through stomata) with absorption of water by the root hairs.
- Writing that Cuscuta makes its own food; it is a **parasite** without chlorophyll.
- Forgetting that iodine test shows **starch** (blue-black), not glucose, and that the leaf should first be boiled in alcohol to remove chlorophyll.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Life Processes in Plants** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** oxygen	By-product.
2	**(a)** parasites	They take food from hosts.
3	**(b)** respiration	All the time.
4	**(d)** shedding leaves	Old leaves and bark.
5	**(a)** leaves	Green parts.
6	**(c)** saprotrophs	Decompose organic matter.
7	**(b)** carbon dioxide	Used up.
8	**(c)** storing in bark and fruits	Excretion in plants.
9	**(d)** stomata	Open and close.
10	**(a)** nitrogen	Gets nitrogen from insects.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Add iodine (1); Blue-black colour (1).	1+1
12	Loss of water vapour from leaves (1); Through stomata (1).	1+1
13	Trap and digest insects (1); Soil is poor in nitrogen (1).	1+1
14	Shed leaves (1); Store wastes in bark (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Takes food from host (1); Cuscuta (1); Harms host (1).	1+1+1
16	For proteins (1); From soil nitrates (1); Fertilisers and bacteria (1).	1+1+1
17	Release oxygen (1); Take in carbon dioxide (1); Keep a balance (1).	1+1+1
	OR: Feeds on dead matter (1); Fungi, mushrooms (1); Decomposers (1).	1+1+1
18	Flat to catch light (1); Chlorophyll (1); Stomata for gases (1).	1+1+1
	OR: Root: carrot (1); Stem: sugarcane (1); Leaf: spinach (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Gas exchange for CO2 (1); Release of oxygen (1); Loss of water vapour (1); Guard cells control (1); Close at night (1).	1+1+1+1+1
	OR: Boil leaf in water (1); Boil in alcohol to remove chlorophyll (1); Wash and spread (1); Add iodine (1); Observe the colour (1).	1+1+1+1+1
20	Cover a potted plant's leaves with a polythene bag (1); Tie around the stem (1); Keep in sunlight (1); Droplets collect inside (1); Water vapour from leaves (1).	1+1+1+1+1
	OR: Compete for light (1); Tall to reach sunlight (1); Broad leaves catch light (1); More photosynthesis (1); Shade-adapted (1).	1+1+1+1+1

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